

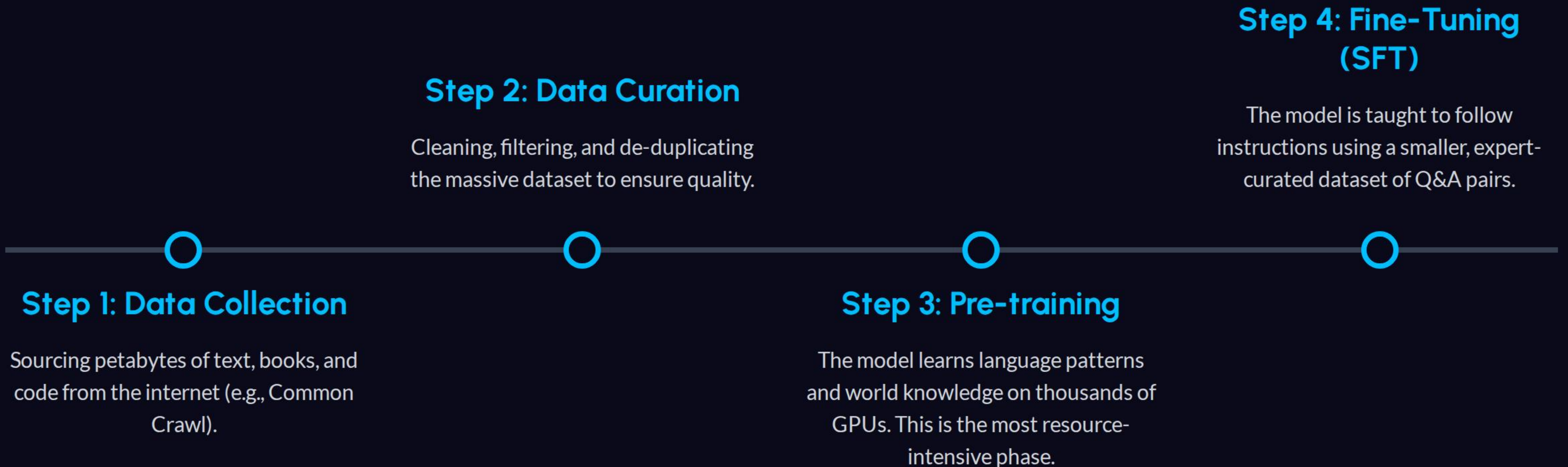
The Anatomy of a Mind

The Training Process & Resource Costs of Generative AI

What Does It Take to Build an LLM?

This presentation breaks down the complex lifecycle and immense resource investment required to create models like GPT, LLaMA, Gemini, and Claude.

The Training Lifecycle: Building the Core



The Training Lifecycle: Alignment & Safety

-  **Step 5: Alignment (RLHF):** Using Reinforcement Learning from Human Feedback (RLHF), the model is "aligned" to be helpful, honest, and harmless.
-  **How RLHF Works:** Human labelers rank multiple model-generated answers from best to worst. A separate "Reward Model" is trained to mimic these human preferences.
-  **Polishing the Model:** The LLM is then trained to generate responses that maximize its "reward score," effectively polishing its behavior.
-  **Evaluation & Deployment:** The final model is tested against academic benchmarks and "red-teamed" for safety vulnerabilities before being deployed.

The 4 Pillars of Training Cost

Training is a multi-dimensional investment. The primary costs are not just financial.

Pillar 1: Compute

>\$100M

Estimated cost for GPT-4's training.

The Engine of AI

This is the single largest expense. Models are trained on massive clusters of specialized GPUs (like NVIDIA H100s), often numbering in the tens of thousands.

Meta's LLaMA 3, for example, was trained on a cluster of 24,000 H100 GPUs, representing a colossal investment in hardware and cloud computing time.

Pillar 2: Data & Human Labor



Raw Data

While web scrapes (like Common Crawl) are low-cost, storing and processing petabytes of data is a significant infrastructure challenge.



Human Labelers

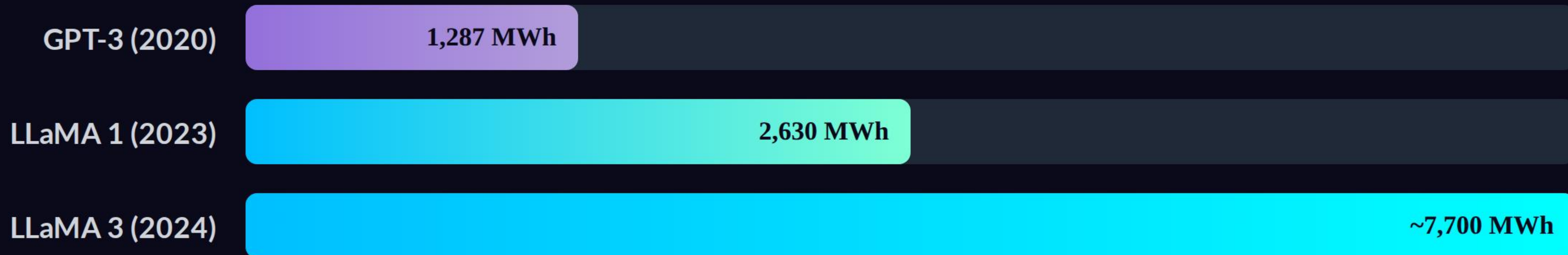
The **real** data cost. Thousands of hours from paid contractors are needed to create the high-quality SFT and RLHF data for alignment.



Licensing

Acquiring rights to high-quality, proprietary datasets (like books, academic articles, and specialized code) can cost millions.

Pillar 3: Energy Consumption



Training consumes vast amounts of electricity for compute and cooling. This has both a high financial cost and a significant environmental footprint.

Pillar 4: Time & Talent

- **Development Cycle:** A state-of-the-art model can take 1-2 years from conception to deployment.
- **Training Runs:** The final pre-training run alone can take several months, running 24/7 without interruption.
- **Human Talent:** The cost of salaries for hundreds of PhD-level AI researchers, engineers, and ethicists is a massive operational expense.

 A team of data scientists collaborating in a modern office.

Putting It All Together: Model Examples

Model	Developer	Est. Compute Cost	Key Feature
GPT-4	OpenAI	>\$100 Million (est.)	Pioneer in large-scale performance and RLHF.
LLaMA 3	Meta	Trained on 24k H100s	Focused on open-source and efficient architecture.
Gemini 1.5	Google	Not Disclosed	Natively multimodal, 1M token context window.
Claude 3	Anthropic	Not Disclosed	Focused on "Constitutional AI" for safety.

Key Takeaways

1. Training is a 5-step lifecycle: Collection → Pre-training → SFT → RLHF → Evaluation.
2. Compute is the dominant financial cost, running into the hundreds of millions.
3. Human labor (for data curation) and Time (for R&D) are critical, costly resources.

Questions?

Thank you.

This artifact is part of my professional portfolio.

Image Sources



<https://resonanzcapital.com/hubfs/AI-Generated%20Media/Images/The%20image%20depicts%20a%20bustling%20modern%20office%20environment%20where%20teams%20of%20analysts%20and%20data%20scientists%20collaborate%20In%20the%20foreground%20a%20diverse%20group%20of%20professionals%20huddle%20around%20a%20large%20screen%20displaying%20dynamic%20graphs%20and%20data%20visualizations%20related%20t-1.jpeg>

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